

EXPRESSIONS in C

1. Summary

This paper presents the notion of an expression in the C/C++ programming language. It also presents the operators and the way an expression is evaluated.

2. Brief theory reminder

2.1. Defining an expression

An **expression** is composed of a single operand or more **operands** glued together by **operators**.

An operand may be:

- A constant.
- A symbolic constant.
- A name for a scalar variable.
- A name for an array variable.
- A name for a structure.
- A name for a type.
- An indexed variable.
- A name for a function.
- A reference to an element of a structure.
- A function call.
- An expression surrounded by parenthesis.

An **operand** has a **value** and a **type** associated with it.

Operators may be unary or binary.

When evaluating an expression, the following are taken into account:

- Operator precedence.
- How operators of the same precedence associate.
- Default conversion rules.

2.2. Operators

The operators of C/C++ belong to one of the following categories:

- Arithmetic operators:
 - Unary operators: **+**, **-**
 - Binary multiplicative operators: *****, **/**, **%**
 - Binary additive operators: **+**, **-**
- Relational operators: **<**, **<=**, **>**, **>=**
- Equality operators: **=**, **!=**
- Logic operators: **!**, **&&**, **||**
- Logic operators on bits: **~**, **<<**, **>>**, **&**, **^**, **|**
- Assignment operators: **=**, **/=**, ***=**, **%=**, **+=**, **-=**, **<<=**, **>>=**, **&=**, **^=**, **|=**
- Pre- and post increment operators : **++**, **—**
- Type cast operators: **(type)**

- Dimension operators: **sizeof**
- Referencing (address) operators: **&**
- Parenthesis operators: **(), []**
- Conditional operators: **?, :**
- Comma operator: **,**
- Dereferencing operator: *****
- Structure component access operator: **., ->**

Note: C++ has a number of additional operators:

- Resolution operator: **::**
- An operator for the reference type: **&**
- An operator to allocate/deallocate the heap:
- Type constructor/destructor invocation operators: **new/delete**

Operator precedence, in the decreasing order of their priority, is given in the table below:

Priority	Operators
15-maximum	() [] • ->
14	+(unary) -(unary) &(unary) *(unary) ++ -- (type) sizeof ! ~
13	*(binary) / %
12	+(binary) -(binary)
11	<< >>
10	< <= > >=
9	= = !=
8	&(binary)
7	^
6	
5	&&
4	
3	? :
2	= <<= >>= += -= *= /= %= &= ^= =
1-minimum	,

Operators on the same row have identical precedence.

Operators are **associated** in the left-to-right order, except for the unary, conditional, and assignment operators which associate right-to-left.

2.3. Default conversion rules

The default conversion rules are stated as follows:

- If a binary operator is applied to operands of the same type, then the result will take the identical type of the operands.
- If a binary operator is applied to operands of the different types, then the operand of an inferior (more restrictive) type is converted by default to the superior type of the other operand, and the result gets the superior type.

Type priority, in decreasing order, is the following:

- long double (maximum priority);
- double;
- float;
- unsigned long;
- long;
- int (minimum priority).

3. Assignments

- 3.1. Write a program to calculate the value $z = x^{**}y$, for variables x and y of the type double.
- 3.2. Explain the difference between real (floating point) division and the integer division.
- 3.3. Write a program to read the value of an angle expressed in degrees and calculates the values for its sine, cosine and tangent.
- 3.4. Write a program to read a positive integer number in the range [1600, 4900]. Knowing that that number represents a year, check whether that year is bissextile or not.
- 3.5. Using conditional expressions, write a program which reads a real value for x and then computes the value for the function::

$$f(x) = \begin{cases} x^2 + 4x + 4 & \text{if } x < -2 \\ 0 & \text{if } x = 0 \\ x^2 + 5x & \text{if } x > -2 \end{cases}$$

- 3.6. Write a program to read a real number x , representing a measurement for an angle in Radians, and then converts it to degrees, minutes, and seconds.
- 3.7. Write a program to simulate the operation of a counting clock (indicating the hour, minute, and second).
- 3.8. Write a program to show the number of bytes the C/C++ data types take in the computer memory.
- 3.9. Convert to binary, by computation, your birth year and the current year.
- a) Show how the two years are represented as data of type int.
 - b) Show the effect the shift operations: left shift by 4 bits, right shift by 2 bits, and 1's complement applied to this data.
 - c) Show the effect of the bitwise operations $\&$, $\wedge$, $|$, upon the provided data.
 - d) Write a program to check the correctness your calculations.
- 3.10. Write a program which effects arithmetic operations upon two data items, one of which is an integer, and the other a real number. Execute this program using values which yield results outside the internal representation limits. What happens when limits are over(under)flowed?
- 3.11. Write a program which reads the integer numbers a, b, c, d and outputs the highest value of fractions a/b and c/d .

3.12. Establish, in a program, the position of a straight line in relationship with a given circle. Your program will read: the coordinates of the center of the circle, its radius, and the coordinates for two points located on the straight line.

3.13. Write a program which reads the planar coordinates of the vertices of a triangle, and then describes the relative position (i.e. above, below, left, right) of this triangle in relationship with a point in the same plane, given by its coordinates.

3.14. Write a program to calculate the ideal weight of a person using the following formulas:

$$W_{\text{male}} = 50 + 0.75 * (\text{height} - 150) + (\text{age} - 20) / 4;$$

$$W_{\text{female}} = W_{\text{male}} - 10;$$

Sex, height (in centimeters), and age (in years) are to be read.

3.15. Write a program to convert Cartesian coordinates of a given point to polar coordinates.